

Experiment (5)

DYNAMICS OF CIRCULAR MOTION “CENTRIPETAL FORCE”

5.1 Introduction

When we are riding a car, bus or train and it rounds a corner, we have a sensation that we are being thrown outside the circular path. There is no obvious force, in the sense that we use in Newtonian mechanics, to explain this sensation. Yet Newton’s laws of motion give us all that we need to understand this circular motion, *if we stick to the inertial reference frame*.

Consider the following questions. Why would a mass rotating in a circular path with *constant speed* be accelerating? What is the net force acting on this mass causing this acceleration? Would it be directed inwards or outwards?

5.2 Objectives

- Learn about the parameters of the uniform circular motion of a body (centripetal force, tangential velocity, angular velocity, centripetal acceleration).
- Investigate the relation between the centripetal force and the mass of the body.
- Investigate the relation between the centripetal force and the angular velocity of the rotating mass.
- Investigate the relation between the centripetal force and the radius of the circular path.

5.3 Theory

The centripetal force is required to keep an object moving in a circular path. For example, the gravity is the centripetal force that keeps satellites orbiting Earth.

When a particle of mass m rotates in a circular path of radius R with a constant speed v (tangent to the path), it is found that the particle acceleration a_{rad} is always directed from the body towards the center of the circle (i.e. the radial direction towards the center) and is given by:

$$a_{rad} = \frac{v^2}{R} = \omega^2 R \quad (5.1)$$

Where ω is the angular velocity and is equal to, $\omega = \frac{v}{R}$

This acceleration is called the ‘centripetal’ acceleration, the word centripetal means “towards the center”.

If the time of one complete period is T , then,

$$\omega = \frac{2\pi}{T},$$

and

$$a_{rad} = \omega^2 R = \frac{4\pi^2 R}{T^2}$$

According to Newton’s second law, the net centripetal force F_c (which is also in the radial direction towards the center) is given by:

$$F_c = \frac{mv^2}{R} = m\omega^2 R \quad (5.2)$$

5.4 Setup

The experimental setup is arranged as shown in Figure 5.1.

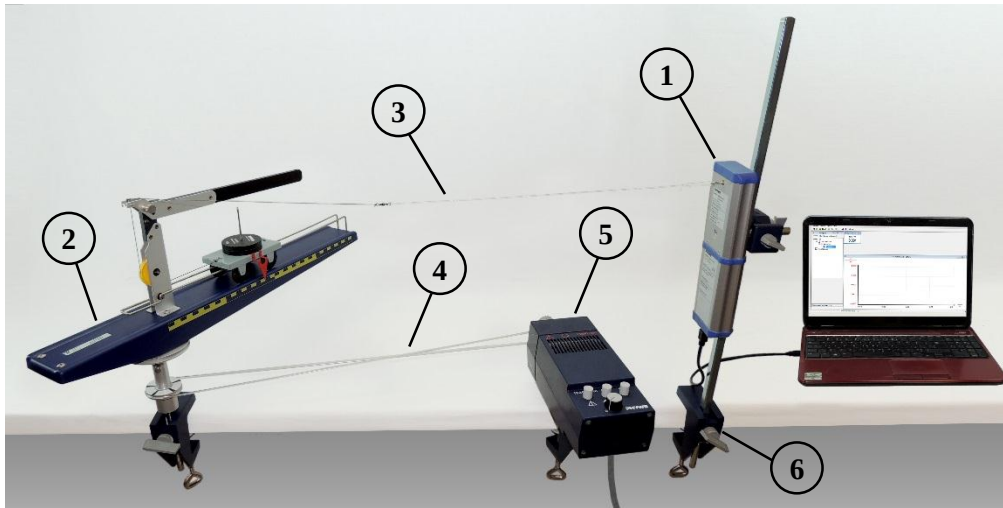


Figure 5.1: Experimental setup for measuring the centripetal force.

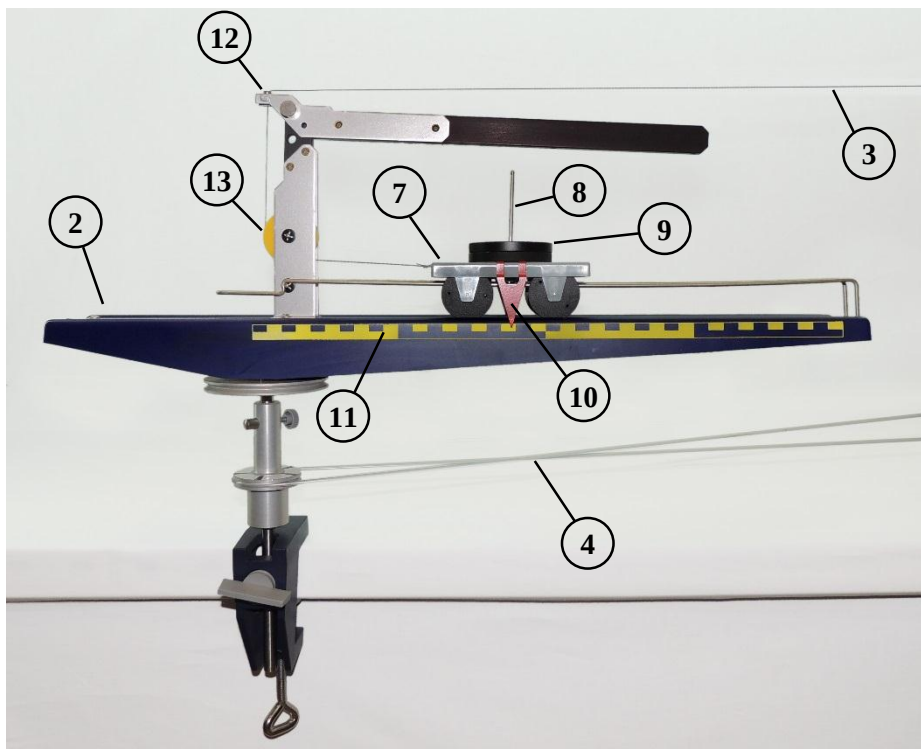


Figure 5.2: Rotating track with movable cart.



Figure 5.3: Two-direction motor with adjustable rpm.



Figure 5.4 : Cobra4 force sensor unit (4N Max.).

- 1 Cobra 4 Force Sensor-Unit
- 2 Rotating cart track
- 3 Fishing line
- 4 Belt
- 5 Laboratory motor
- 6 Force sensor clamp
- 7 Movable cart
- 8 Weight holder
- 9 Slotted Masses
- 10 Radius pointer

- 11 Radius ruler
- 12 The eye
- 13 Fishing line pulley
- 14 Motor shaft
- 15 13000 rpm (Max. speed) button
- 16 Clockwise rotation button
- 17 Counter-clockwise rotation button
- 18 Rpm rotary switch
- 19 Force sensor hook
- 20 Force sensor specifications label

5.5 Procedure

5.5.1 Task (1): To address our second objective, which is to determine the relation between the centripetal force and the mass, we will measure the variation of the centripetal force as a function of the mass of the rotating object.

- 1- Start the measure software package on the PC. All necessary recording settings will start automatically as shown in **Error! Reference source not found..**

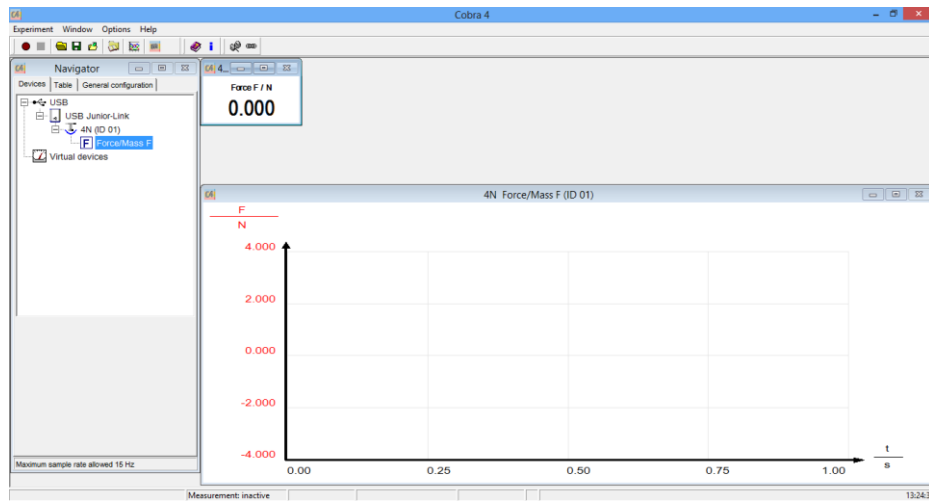


Figure 5.5: Measure software startup window.

- 2- Place 400 gm of slotted mass (9) on the weight holder (8) of the cart (7) as shown in Figure 5.6.
- 3- Adjust the radius of the circular path taken by the mass to $r = 20\text{ cm}$. This radius should remain constant throughout this task.

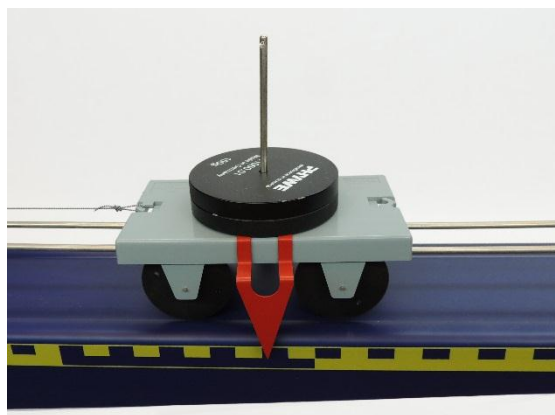


Figure 5.6: Movable cart with added slotted masses.



The radius of the circular path can be adjusted by loosening the clamp (6) and moving the support rod holding the force sensor (1) back and forth until the required radius is adjusted then fastening the clamp again as shown in Figure 5.7. Make use of the red radius pointer (10) on the side of the cart (7) and the radius ruler (11) on the track to adjust the correct radius. $r = 20\text{ cm}$ adjustment is shown in Figure 5.8.

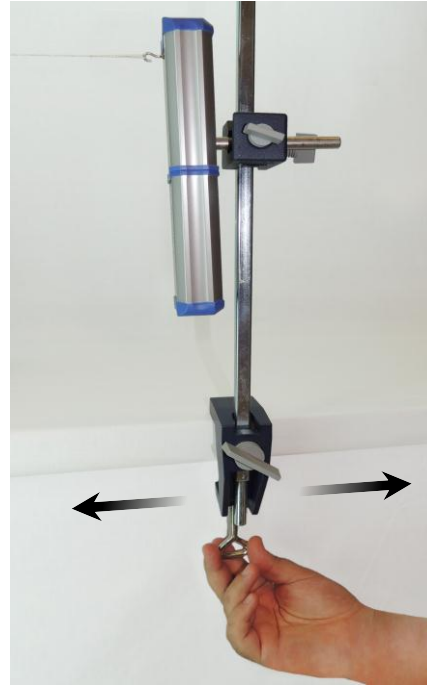


Figure 5.7: Loosening and fastening the sensor clamp to enable moving the support rod back and forth.

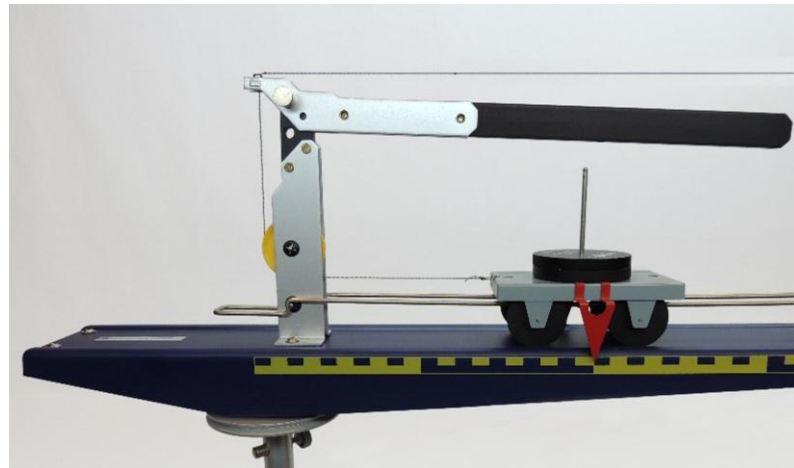


Figure 5.8: The $r = 20\text{ cm}$ adjustment.



Make sure that the fishing line (3) connected to the force sensor hook (19) and passing through the eye (12) is horizontal (i.e. parallel to the cart track (2) and perpendicular to the force sensor (1)) as shown in Figure 5.9. Failing to adjust the fishing line correctly will result in measuring only a component of the force not the total force.

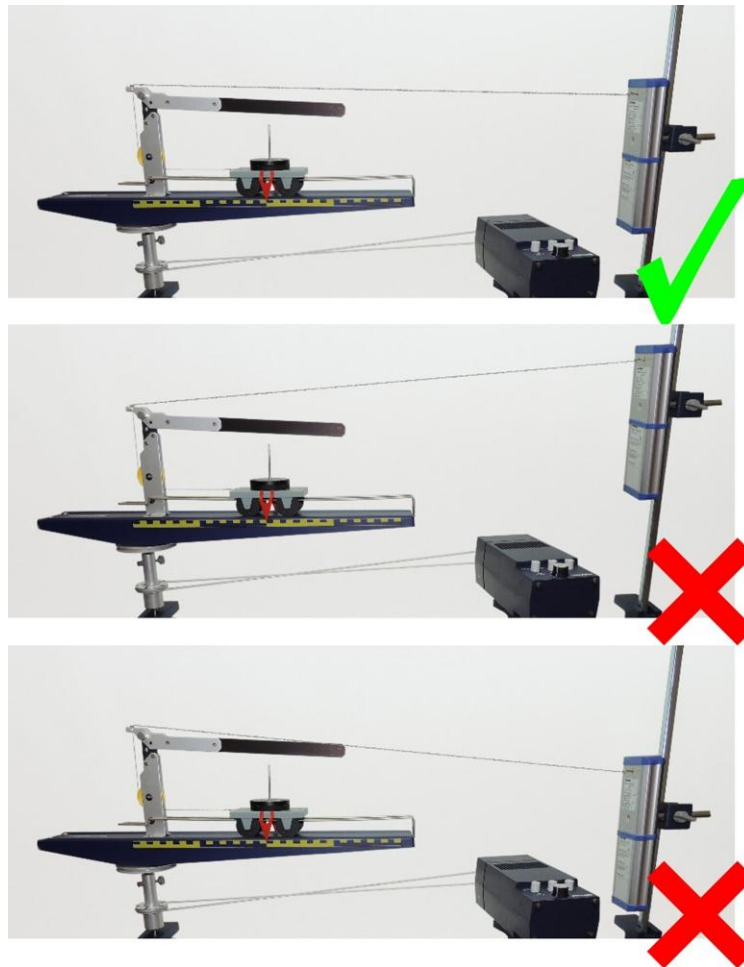


Figure 5.9: Correct and incorrect fishing line adjustment.

- 4- Read the following safety precautions ⚠ carefully before carrying out this step and ask your Professor or TA if you do not understand any of these points.



- **Before switching on the motor**, make sure all units, which are to be driven, **are firmly fastened** and that their holding devices are safely mounted.
- Never use the **“13000 rpm”** (15) button.
- Put on your **protective glasses** before switching on the motor
- The **rotation should not be hampered or blocked** for long periods, as this will damage the motor due to the increasing current passing through its coils.
- **No one is to get close enough to collide** with the apparatus as it rotates. Avoid placing your head near the level of the rotating apparatus.
- **Hold the fishing line up** (see Figure 5.10) before the motor starts rotating and let it go after the cart track completes one rotation.

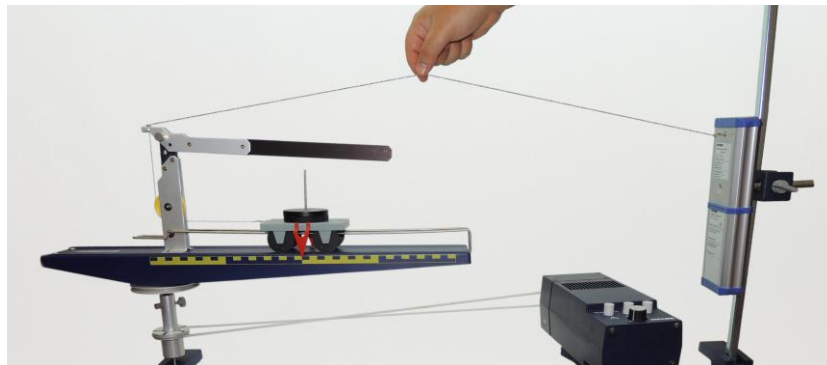


Figure 5.10: Holding the fishing line up before the motor starts rotating.

- **If the fishing line is left loose** when the motor starts rotating, it - most probably- will **wind around the axis** of the apparatus, if this occurs, **switch off the motor (5) or unplug it** from the wall socket at once and ask a TA for help.

Turn the laboratory motor (5) on by pressing button (16) or (17) then use the rpm rotary switch (18) to adjust the rpm (revolutions per minute) of the motor, which in turn rotates the cart track (2).



It will be noticed that the cart track will not rotate with the expected angular velocity instantly; one should wait until the angular velocity of the cart track seems to be constant before moving on to the next step.



The centripetal force measured by the force sensor (1) depends on the angular velocity of the rotating mass. As shown in Figure 5.12, the force sensor can't measure any force exceeding 4. That is why, on adjusting the motor angular velocity, one should check the measured force window shown in Figure 5.11 on the measure software to make sure that the force doesn't exceed 4N for the adjusted angular velocity of the motor.

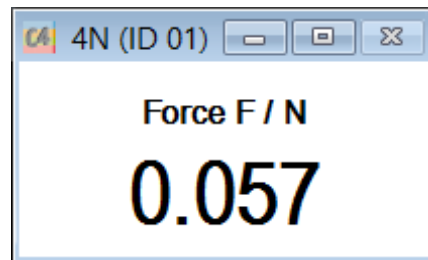


Figure 5.11: Measured force window.

- 5- Start recording the measured force in the measure software by clicking the “Record” button (●) found in the software toolbar shown in Figure 5.12. The measured values by the force sensor are now being recorded.

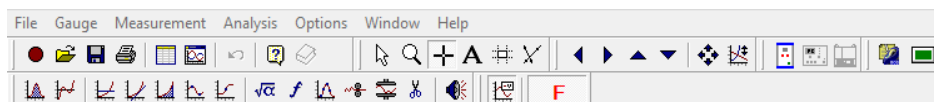


Figure 5.12: Measure software toolbars.

- 6- After approximately 10 full rotations, you can terminate the recording from the “Stop” button (■), the output graph will appear automatically.



The output force vs. time graph will have a sinusoidal behavior although - theoretically - the measured force should be constant with time. This can be explained as follows : When the cart track (2) is rotating, the eye (12) moves slightly back and forth when the apparatus is viewed from the side, this in turn causes the circular path radius to change slightly (about the adjusted radius) during the circular motion (Recall that $F \propto R$ at certain ω). This eccentricity (deviation from the center) of the eye is intentional as it allows the determination of the angular velocity of the cart track as will be shown later.

- 7- Press the motor button which was previously pressed in step 4 (button (16) or (17)) to stop the motor without touching the rpm rotary switch.
- 8- Use the “Survey tool” () to mark a number of cycles on the graph, count the cycles lying between the two vertical markings on the graph as shown in Figure 5.13.

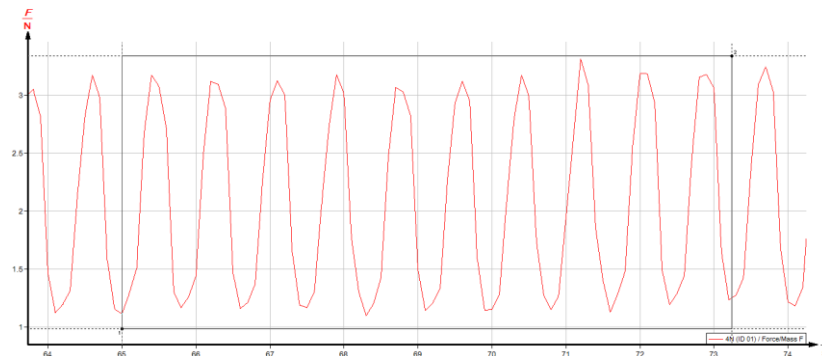


Figure 5.13: Marking 10 cycles using the “survey tool”.

The Δx in the “Survey tool” window shown in Figure 5.14 is the time taken to complete the cycles bounded by the vertical markings on the graph (note that the time is on the horizontal x-axis). Calculate the periodic time $= \frac{\Delta x}{N}$, where N is the number of cycles lying between the two vertical markings.

Calculate the angular velocity ω from the periodic time, where
 $(\omega = \frac{2\pi}{T})$.

X1: 65.00 s	X2: 73.24 s	ΔX : 8.24 s
Y1: 0.985 N	Y2: 3.337 N	ΔY : 2.352 N

Figure 5.14: Survey tool window.



The force vs. time graph has the force on the vertical y-axis and the time on the horizontal x-axis, that's why the Δx in the Survey tool window is the difference in time bounded by the vertical markings on the graph.

- 9- Use the “Mark” tool (+) to select three or four periods from the graph then press the “Show average value” tool (⏏), the average value of the force during these selected periods will appear in a dialogue box as shown in Figure 5.15.
- 10- Repeat steps 4 to 9 several times for error analysis with 400 gm.
- 11- Repeat steps 4 to 9 several time for each of the following masses : 350 gm, 300 gm and 150 gm.

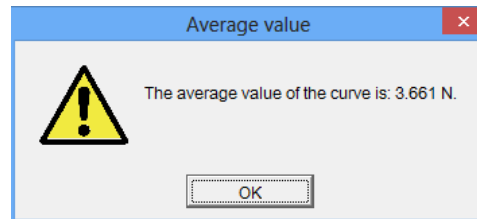


Figure 5.15: Average value dialogue box.



On repeating steps 4 to 9, Make sure not to change the motor's rpm because the angular velocity is required to be almost constant through out this task.

- 12- Plot the relation between the mass of the rotating object (taking the mass of the cart into consideration) and the centripetal force acting on it. Find the slope of the straight line describing the relation and compare it to the value of $\omega^2 r$.

5.5.2 Task (2): Verifying the relation between the centripetal force and the radius of the circular path.

- 1- Place a 150 gm mass onto the cart. This mass should remain constant through out this task.
- 2- Adjust the radius of the circular path taken by the mass to $r = 30 \text{ cm}$.
- 3- Repeat steps 4- to 0 from Task (1) several times for each of the following values for the radius of the circular path taken by the mass: $r = 30 \text{ cm}, 25 \text{ cm}, 20 \text{ cm}$ and 15 cm .



Before recording in step 5 (from Task (1)), make sure that the force sensor reads a value less than 4N when the radius of the circular path $r = 30 \text{ cm}$. If the force sensor is reading more than 4 N, decrease the motor's rpm until reaching a force reading within the range.



On repeating the steps 4 to 9 (from Task (1)), make sure not to change the motor's rpm because the angular velocity is required to be almost constant through out this task.

- 4- Plot the relation between the radius of the circular path taken by the mass of the rotating object and the centripetal force acting on it. Find the slope of the straight line describing the relation and compare it to the value of $m\omega^2$.

5.5.3 Task (3): Variation of the centripetal force as a function of the angular velocity of the rotating mass.

- 1- Place 150 gm mass onto the cart. This mass should remain constant through out this task.
- 2- Adjust the radius of the circular path taken by the mass to $r = 20\text{ cm}$. This radius should also remain constant through out this task.
- 3- Repeat steps 4- to 0 from Task (1) but in step 4 set the motor's rpm to four different values which will result in four different values for the angular velocity (the operating angular velocity is to be calculated in step 8-). For each adjusted rpm, repeat the steps several times for error analysis without changing the motor's rpm for error analysis.



Before recording in step 5 Task (1)), make sure that the force sensor reads a value less than 4N for each different rpm . If the force sensor is reading more than 4N, decrease the motor's rpm until reaching a force reading within the range.

- 4- Plot the relation between the square of the angular velocity of the mass of the rotating object and the centripetal force acting on it. Find the slope of the straight line describing the relation and compare it to the value of mr .